

1. The Box Model (Spacing & Sizing)

These are the most fundamental properties for structuring your elements.

- **width**: Sets the width (e.g., `100%`, `500px`).
- **max-width**: Limits width so content doesn't get too wide on large screens (e.g., `1200px`).
- **height**: Sets fixed height. (`10vh` means 10% of the vertical height)
- **min-height**: **Crucial for Hero sections**. `min-height: 100vh` makes a section take up the full screen height.
- **box-sizing**: Always set `box-sizing: border-box;` in your reset (*). It ensures padding doesn't increase the total width of an element.

Multi-Part Values: Margin & Padding

Margin: gap between two elements.

Padding: gap inside an element.

Can be: top, bottom, left, right.

Both **margin** (outer space) and **padding** (inner space) follow the **Clockwise Rule** (Top, Right, Bottom, Left).

- **1 Value**: `padding: 20px;` (Apply 20px to **all 4 sides**)
 - **2 Values**: `padding: 20px 50px;` (20px **Top/Bottom**, 50px **Left/Right**)
 - **3 Values**: `margin: 10px 20px 30px;` (Top, Left/Right, Bottom)
 - **4 Values**: `margin: 10px 20px 30px 40px;` (Top, Right, Bottom, Left)
-

2. Visual Styling (Backgrounds & Borders)

- `background-color`: Sets solid colors (e.g., `#ECEFF4`).
- `border-radius`: Rounds corners.
 - `border-radius: 50%` makes a perfect circle (for profile pictures).
 - `border-radius: 8px` gives modern soft corners.
- `cursor: cursor: pointer;` changes the mouse arrow to a hand (used for buttons/links).

Multi-Part Value: Border

The `border` property is a shorthand for three properties: width, style, and color.

```
/* Order: width | style | color */  
border: 1px solid #000;
```

Multi-Part Value: Box Shadow

This creates depth/elevation (like the white cards in your screenshots).

```
/* Order: x-offset | y-offset | blur-radius | spread | color */  
box-shadow: 0px 4px 15px rgba(0, 0, 0, 0.1);
```

- **0px (X)**: Shadow doesn't move left or right.
 - **4px (Y)**: Shadow moves down 4px.
 - **15px (Blur)**: How soft/blurry the shadow is.
 - **Color**: Usually black with low opacity (rgba).
-

3. Typography (Text Styling)

- `font-family`: The font used (e.g., `'Arial'`, `sans-serif`).
 - `font-size`: Size of text (`16px` or `1rem`).
 - `font-weight`: Thickness (`400` for normal, `700` for bold).
 - `color`: Text color.
 - `text-align`: `left`, `center`, or `right`.
 - `text-decoration`: `text-decoration: none`; removes the underline from links (`<a>`).
 - `line-height`: Vertical space between lines of text (makes paragraphs readable).
-

4. Flexbox Properties

Most of your exam questions (Navbars, Split screens) are best solved with Flexbox.

Properties for the Parent (Container):

- `display: flex`; Activates flexbox.
- `flex-direction`:
 - `row` (Default): Items side-by-side (Left to Right).
 - `column`: Items stacked (Top to Bottom).
- `justify-content` (Moves items along the Main Axis):
 - `space-between`: Pushes items to edges (Perfect for Navbars: Logo left, Links right).
 - `center`: Centers content horizontally.
 - `flex-start` / `flex-end`.
- `align-items` (Moves items along the Cross Axis):
 - `center`: Vertically centers items (e.g., text inside a button).
 - `stretch`: Makes items the same height.
- `gap`: Adds space between flex items (e.g., `gap: 20px`).

Properties for the Children (Items):

- `flex`: Shorthand for growing/shrinking.
 - `flex: 1`; Makes the item take up all available remaining space. (Used in your split-screen: Left Panel `flex: 1`, Right Panel `flex: 1`).
-

5. CSS Grid Properties

Best used for 2-dimensional layouts, like the **Dashboard** or photo galleries.

Properties for the Parent:

- `display: grid;` Activates grid.
- `grid-template-columns`: Defines width of columns.
 - `repeat(3, 1fr)`: Creates 3 equal columns.
 - `200px 1fr`: Sidebar is 200px, main content is the rest.
- `gap`: Space between rows and columns.

Properties for the Children:

- `grid-column`: Defines how many columns an item spans.
 - `grid-column: 1 / -1;` Spans the full width.
 - `grid-column: span 2;` Takes up 2 slots.
-

6. Positioning (For specific details)

Used for small details like the "Eye" icon inside a password input, or a "notification badge."

- `position: relative;` The parent box.
- `position: absolute;` The item you want to move freely inside the parent.
- `top, right, bottom, left`: Coordinates to move the absolute item.
- `z-index`: Layers elements (higher number = on top).

Example (Password Eye Icon):

```
.input-group {  
  position: relative; /* Parent */  
}  
.eye-icon {  
  position: absolute; /* Child */  
  right: 10px;      /* Stick to right */  
  top: 50%;         /* Move down 50% */  
  transform: translateY(-50%); /* Perfect vertical center */  
}
```